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Department of Artificial Intelligence and Data Science

24ADI204 - DATA SCIENCE AND VISUALISATION

DATA-VIZ HACK-A-BIT (2025-2026)

"Data-Driven IPL Insights" A Moneyball Approach to Player Valuation

A Project Report submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled "IPL Data Analysis and Visualization" is the Bonafide work of the following students of the Department of Artificial Intelligence and Data Science, Kumaraguru College of Technology, who carried out this project under the supervision of the faculty mentioned below.

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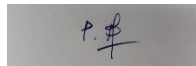
Internal Examiner

External Examiner

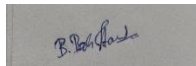
DECLARATION

We affirm that the project work titled "IPL Data Analysis and Visualization", submitted in partial fulfillment for the award of the degree of B.Tech in Artificial Intelligence and Data Science, is the original work carried out by us. This work has not formed part of any other project submitted for the award of any degree or diploma, either in this or any other University.

24BAD021 - Dinesh Kumar P



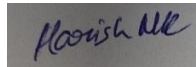
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Abstract

The Indian Premier League (IPL) is one of the most financially dynamic sporting events in the world, where franchises invest crores of rupees during player auctions, often guided by instinct and reputation rather than hard data. This project applies a Moneyball-style data science framework to the IPL 2025 Auction Dataset, which contains 574 player records across 21 attributes including age, nationality, role, international caps, IPL experience, and reserve price.

The project followed a complete data science pipeline: raw data ingestion, data quality assessment, cleaning and imputation, exploratory data analysis, feature engineering, dimensionality reduction via Principal Component Analysis (PCA), and interactive dashboard construction.

Key Insight: Key findings: capped players command significantly higher base prices, overseas players carry a nationality premium, and the Prime age bracket (25-30 years) represents the auction's sweet spot for value.

The analysis culminated in a Plotly-based interactive dashboard that narrates findings as a coherent visual story, enabling franchise decision-makers to make smarter, budget-conscious auction choices.

Chapter 1: Introduction

1.1 Background

The Indian Premier League was established in 2008 and has since grown into the world's most watched cricket league and one of its most valuable sports properties. Historically, the player acquisition process has been influenced by name recognition, recent tournament form, and subjective scouting — factors that do not always translate into on-field value.

The Moneyball philosophy, popularised by the Oakland Athletics baseball team and Michael Lewis's book of the same name, demonstrated that statistical analysis could reveal undervalued players whom traditional scouting overlooked. This project applies that same philosophy to cricket: treating the IPL auction as a market with exploitable inefficiencies that data can expose.

1.2 Objectives

- Perform an end-to-end data science workflow on the IPL 2025 Auction dataset.
- Clean and preprocess raw, messy spreadsheet data including missing values, inconsistent formats, and type errors.
- Conduct thorough Exploratory Data Analysis (EDA) to uncover patterns in player valuation.
- Engineer meaningful features through scaling, encoding, and log transformation.
- Apply Principal Component Analysis (PCA) to reduce dimensionality and identify feature interactions.
- Build an interactive visual dashboard that communicates insights in a story-driven format.

- Derive actionable recommendations for franchise decision-makers on budget-efficient player selection.

1.3 Scope

This project covers the full preprocessing, EDA, feature engineering, dimensionality reduction, and dashboard construction pipeline. The dataset is the Tata IPL Auction List 2025 from Kaggle. Findings are interpreted within the context of IPL franchise strategy and do not constitute financial advice.

1.4 Alignment with Sustainable Development Goals

- SDG 9 - Industry, Innovation and Infrastructure: Promotes data-driven decision-making in a major industry.
- SDG 8 - Decent Work and Economic Growth: Encourages fair, performance-based valuation of players.
- SDG 12 - Responsible Consumption and Production: Enables franchises to allocate auction budgets responsibly.

Chapter 2: Problem Statement

Core Question: Does a player's age, role, nationality, or capped status genuinely influence their base auction price — and can data identify undervalued players who deliver strong output within limited budgets?

IPL franchises operate under a fixed auction budget. Bidding wars for high-profile names often leave teams cash-strapped, limiting their ability to build a balanced squad. Conversely, several talented players — particularly young domestic cricketers and uncapped internationals — go unnoticed or are acquired cheaply despite impressive underlying statistics.

This project addresses three specific research questions:

- Which player attributes (age, role, nationality, cap status, IPL experience) most strongly predict reserve auction price?
- Is there a systematic nationality or experience premium in how players are priced, and is it statistically justified?
- Who are the genuinely undervalued players: those with strong performance credentials priced below their expected market value?

Chapter 3: Dataset Description

3.1 Source

The dataset is the Tata IPL Auction List 2025, sourced from Kaggle at: <https://www.kaggle.com/datasets/bhadramohit/tata-ipl-auction-list2025-dataset>. It was selected for being raw and uncleaned — a requirement of the DATA-VIZ HACK-A-BIT guidelines — and for containing sufficient attributes and records to support a meaningful multi-stage analysis.

3.2 Dataset Dimensions

Property	Value
Total Records (after cleaning)	574 players
Total Columns (raw)	21 columns
Player Roles	Batter, Bowler, All-Rounder, Wicketkeeper
Nationalities	13 countries (India dominant at 366 players)
Target Variable	Reserve_Price (in Lakhs)
Data Format	CSV (.csv) — spreadsheet export with messy headers

3.3 Feature Description

Column	Type	Description
First_Name / Surname	Text	Player full name — identifier only
Country	Categorical	Nationality — 13 unique countries
Age	Numeric	Age in years (range: 14 to 42, mean: 27.3)
Role / Specialism	Categorical	Batter, Bowler, All-Rounder, Wicketkeeper
Test_Caps	Numeric	Number of Test match appearances
ODI_Caps	Numeric	One-Day International appearances
T20_Caps	Numeric	T20 International appearances
IPL_Matches	Numeric	Number of IPL matches played
Cap_Status	Categorical	Capped or Uncapped status
Reserve_Price	Numeric	Base auction price in Lakhs (target variable)

3.4 Initial Data Issues Identified

- The first two rows of the CSV were metadata sub-headers from the original spreadsheet export, requiring removal before analysis.
- All numeric columns (Age, caps, IPL_Matches) were stored as object dtype due to header contamination, requiring explicit type casting.
- Reserve_Price was stored as a text string mixing "Cr" and "Lakh" denominations — conversion to a uniform numeric unit (Lakhs) was essential.
- The Cap_Status column had numeric values mistakenly shifted into it due to spreadsheet column misalignment.
- State_Association was missing for all 208 overseas players — a structured missingness pattern requiring conditional imputation.

Chapter 4: Data Preprocessing (Week 3)

Data preprocessing was the most technically demanding phase of this project. The raw dataset carried several structural issues inherited from its spreadsheet origin, requiring methodical resolution before any meaningful analysis could begin.

4.1 Column Renaming and Header Cleanup

All 21 columns were renamed from abbreviated spreadsheet-style labels to clear snake_case identifiers. The first two rows — which contained sub-header text — were dropped, reducing the usable dataset to 574 player records.

```
df.columns=[
    "List_Sr_No", "Set_No", "Set_Code", "First_Name", "Surname",
    "Country", "State_Association", "Date_of_Birth", "Age", "Role",
    "Batting_Style", "Bowling_Style", "Test_Caps", "ODI_Caps",
    "T20_Caps", "IPL_Matches", "Previous_IPL_Teams", "Current_Team",
    "No_of_Matches", "Cap_Status", "Reserve_Price"
]
df = df.drop(index=[0,1]).reset_index(drop=True)
print("Shape After Removing First Two Rows:", df.shape)
# Output: Shape After Removing First Two Rows: (574, 21)
```

4.2 Type Conversion

Numeric columns were stored as Python object strings due to header row contamination. Explicit `pd.to_numeric()` conversion was applied with `errors="coerce"` to handle any residual nonnumeric values.

```
caps_cols =
["Test_Caps", "ODI_Caps", "T20_Caps", "IPL_Matches", "No_of_Matches"]
for col in caps_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
print("Data types after conversion:")
print(df[caps_cols].dtypes)
```

4.3 Reserve Price Parsing

The Reserve_Price column contained mixed-format strings such as "2 Cr" and "30 Lakh". A custom parsing function was written to extract the numeric value and convert everything to a uniform Lakh scale (1 Cr = 100 Lakhs).

```
import re

def convert_price(price):
    if pd.isna(price):
        return np.nan
    price = str(price)
    number = re.findall(r"\d+\.?\.?\d*", price)
    if not number:
        return np.nan
    number = float(number[0])
    if "Cr" in price:
        return number * 100
    elif "Lakh" in price:
        return number

df["Reserve_Price_Numeric"] = df["Reserve_Price"].apply(convert_price)
df["Reserve_Price_Numeric"].fillna(df["Reserve_Price_Numeric"].median(), inplace=True)
print("Reserve Price conversion complete.")
print(df["Reserve_Price_Numeric"].describe())
```

4.4 Cap Status Column Fix

A structural quirk caused numeric values representing "No_of_Matches" to appear in the Cap_Status column for certain rows. These were identified, shifted to the correct column, and Cap_Status entries were standardised to "Capped" or "Uncapped".

```
df["Cap_Status"] = df["Cap_Status"].astype(str)
mask = df["Cap_Status"].str.isnumeric()
df.loc[mask, "No_of_Matches"] = df.loc[mask, "Cap_Status"]
df.loc[mask, "Cap_Status"] = np.nan
df["Cap_Status"] = df["Cap_Status"].str.strip().str.lower()
df["Cap_Status"] = df["Cap_Status"].replace({"capped": "Capped", "uncapped": "Uncapped"})
print("Cap_Status value counts after fix:")
print(df["Cap_Status"].value_counts())
```

4.5 Missing Value Handling Strategy

Column	Missing Count	Strategy	Justification
State_Association	211	Conditional: 'Not Applicable' for overseas	Structured missingness
Test_Caps	441	Median imputation	Skewed distribution
ODI_Caps	370	Median imputation	Right-skewed data

T20_Caps	358	Median imputation	High variance
Reserve_Price	335	Retained as NaN	Unknown for uncontracted players
Cap_Status	181 (post-fix)	Mode imputation	Categorical binary column

4.6 Outlier Detection and Decision

Outlier detection was performed using the Interquartile Range (IQR) method. Outliers were identified in key columns: Reserve Price (87), IPL_Matches (278), T20_Caps (214), ODI_Caps (201), and Test_Caps (132).

```
check_cols =
["Age", "Test_Caps", "ODI_Caps", "T20_Caps",
"IPL_Matches", "Reserve_Price_Numeric"] outlier_counts
= {} for col in check_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    n = len(df[(df[col] < Q1-1.5*IQR) | (df[col] >
Q3+1.5*IQR)])    outlier_counts[col] = n    print(f"{col} ->
Outliers: {n}")
```

Key Insight: Decision: Outliers were RETAINED. In sports analytics, extreme values represent genuine superstars (e.g., Virat Kohli, Jos Buttler) who legitimately command premium prices. Removing them would destroy the very signal being analysed.

4.7 Data Quality Report Summary

Metric	Before Cleaning	After Cleaning
Total Rows	577	574
Duplicate Rows	0	0
Columns with Null Values	16 of 21	Managed
Reserve Price Format	Mixed text (Cr/Lakh)	Uniform numeric (Lakhs)
Cap Status Alignment	Misaligned in source	Corrected
Dtype Correctness	Most columns as object	All numeric correctly cast

Chapter 5: Exploratory Data Analysis (Week 4)

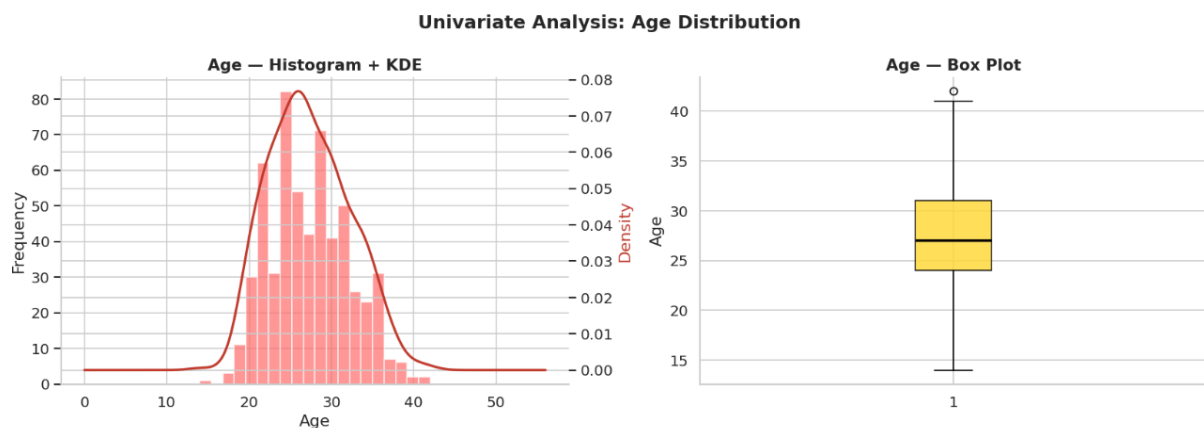
EDA was conducted in two stages: univariate analysis to understand individual feature distributions, and bivariate analysis to examine relationships between player attributes and auction price. All visualizations were produced using Matplotlib, Seaborn, and Plotly.

5.1 Univariate Analysis

5.1.1 Age Distribution

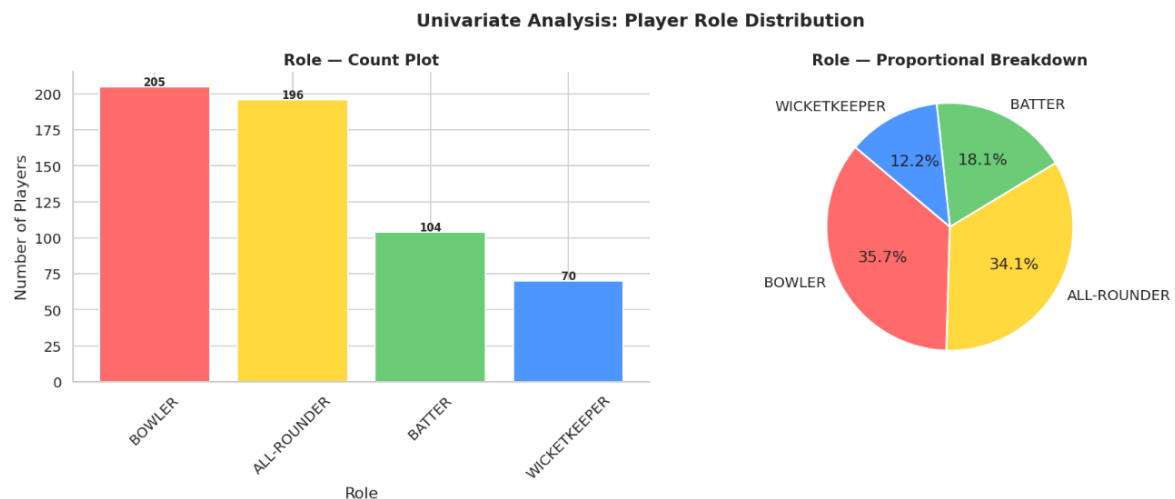
The age distribution is approximately bell-shaped with a slight right skew. Mean age is 27.3 years, median is 27, spanning 14 to 42 years. The highest density falls in the 24-30 age band.

```
df["Age"] = pd.to_numeric(df["Age"], errors='coerce')
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].hist(df["Age"].dropna(), bins=20, color='#FF6B6B',
alpha=0.7, edgecolor='white') ax2 = axes[0].twinx()
df["Age"].dropna().plot.kde(ax=ax2, color='#C0392B', linewidth=2)
ax2.set_ylabel("Density")
bp = axes[1].boxplot(df["Age"].dropna(), patch_artist=True,
medianprops=dict(color='black', linewidth=2))
bp['boxes'][0].set_facecolor('#FFD93D')
plt.suptitle("Univariate Analysis: Age Distribution",
fontsize=15, fontweight='bold') plt.tight_layout() plt.show()
print(f"Mean Age: {df['Age'].mean():.1f} | Median: {df['Age'].median():.1f}")
```



5.1.2 Role Distribution

Batters form the largest category, followed by Bowlers, All-Rounders, and Wicketkeepers. This reflects the IPL's batting-heavy format. Wicketkeepers are the smallest group as each team needs only one or two.

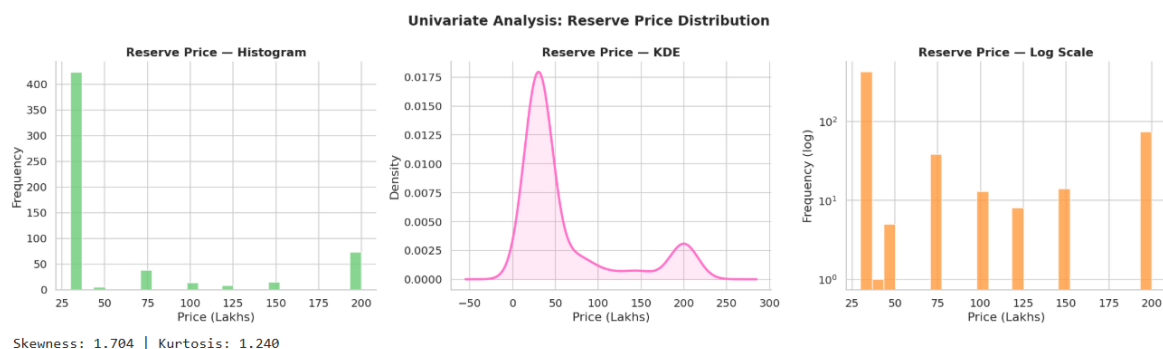


5.1.3 Nationality Distribution

India dominates with 366 players (approximately 64%), followed by England (37), Australia (37), South Africa (31), New Zealand (24), West Indies (22), Sri Lanka (19), and Afghanistan (18).

5.1.4 Reserve Price Distribution

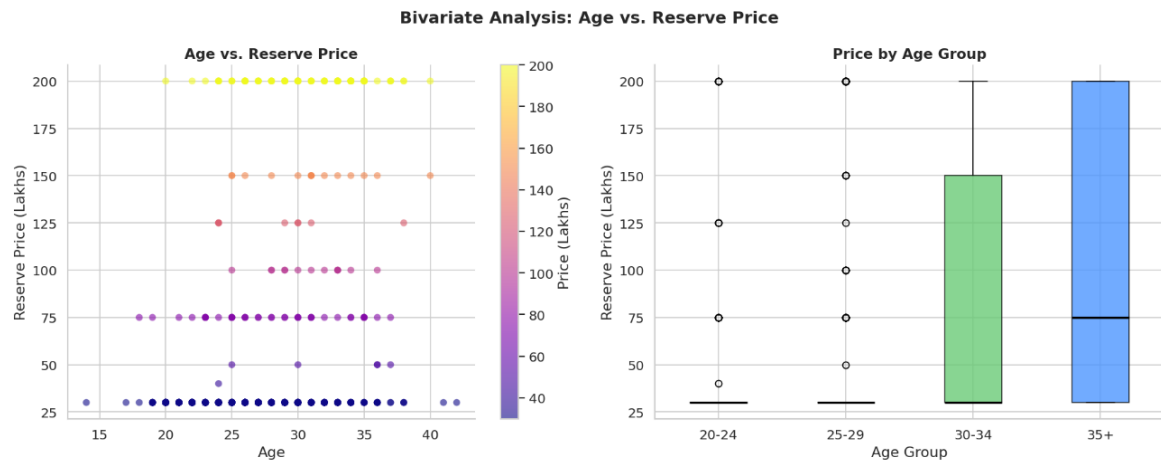
Reserve Price is heavily right-skewed. The majority of players have base prices between 20 and 75 Lakhs, while a small elite group commands 200 Lakhs and above. A log transformation was applied in the feature engineering phase.



5.2 Bivariate Analysis

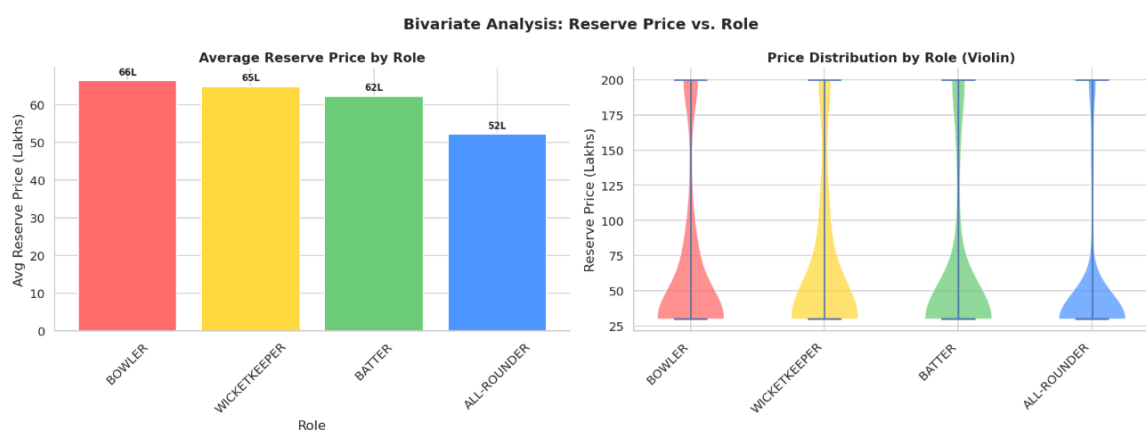
5.2.1 Age vs. Reserve Price

Box plots reveal a clear inverted-U relationship. Players in the Prime age bracket (25-30 years) command the highest median reserve prices. Young and veteran players are generally priced lower.



Age vs. Reserve Price

```
fig, axes = plt.subplots(1, 2, figsize=(15, 6))
# Scatter plot
sc = axes[0].scatter(df["Age"], df["Reserve_Price_Numeric"],
c=df["Reserve_Price_Numeric"], cmap='plasma',
alpha=0.6, s=40, edgecolors='none') plt.colorbar(sc, ax=axes[0],
label='Price (Lakhs)') axes[0].set_title("Age vs. Reserve Price")
# Age group box plot
age_groups = ["20-24", "25-29", "30-34", "35+"] bins
= [20, 25, 30, 35, 50]
df["Age_Band"] = pd.cut(df["Age"], bins=bins, labels=age_groups, right=False)
grp_data = [df[df["Age_Band"]==g]["Reserve_Price_Numeric"].dropna().values
for g in age_groups]
axes[1].boxplot(grp_data, patch_artist=True,
medianprops=dict(color='black',
linewidth=2)) axes[1].set_xticklabels(age_groups)
axes[1].set_title("Price by Age Group")
plt.suptitle("Bivariate Analysis: Age vs. Reserve
Price") plt.tight_layout() plt.show()
```

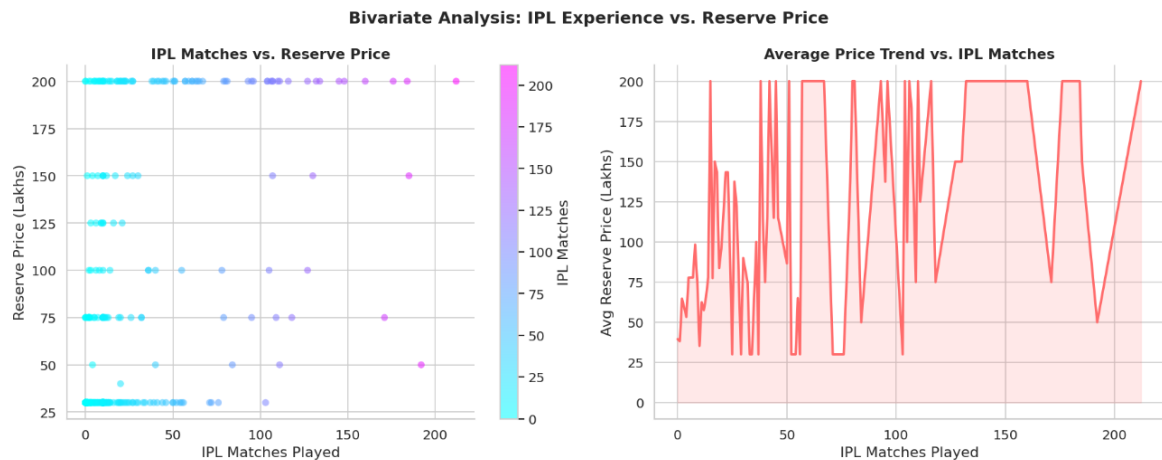


5.2.2 Cap Status vs. Reserve Price

Capped players are priced two to three times higher than Uncapped counterparts on average. The wider IQR for Capped players confirms international recognition as a significant pricing driver.

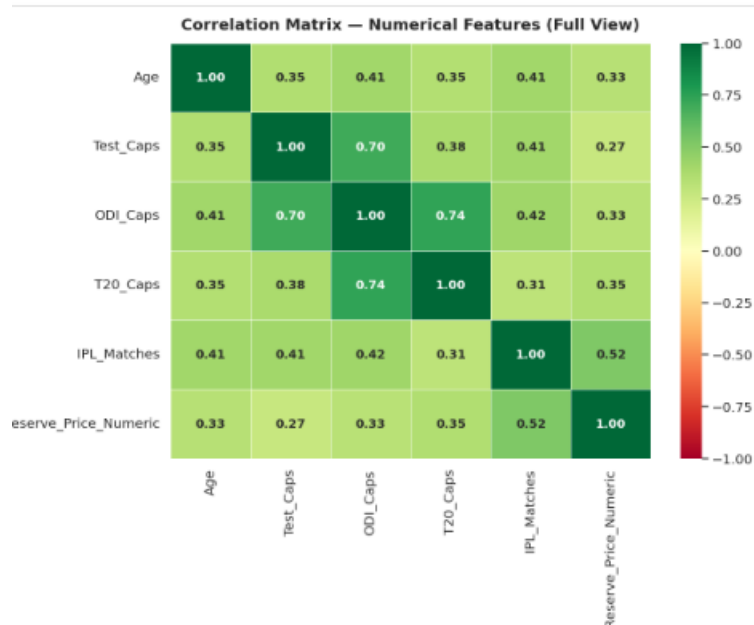
5.2.3 IPL Experience vs. Reserve Price

A positive trend is visible: players with more IPL experience generally command higher base prices. The trend is non-linear — price growth accelerates after approximately 50 matches.



5.2.4 Correlation Heatmap

```
fig, ax = plt.subplots(figsize=(10, 8))
corr = df[["Age", "Test_Caps", "ODI_Caps", "T20_Caps",
          "IPL_Matches", "Reserve_Price_Numeric"]].corr()
mask = np.zeros_like(corr, dtype=bool)
mask[np.triu_indices_from(mask)] = True
sns.heatmap(corr, annot=True, fmt=".2f", cmap="RdYlGn",
            linewidths=0.5, linecolor='white', ax=ax, mask=mask,
            vmin=-1, vmax=1, annot_kws={"size":11, "weight": "bold"})
ax.set_title("Correlation Matrix - Numerical Features",
            fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



Key Insight: *IPL_Matches* has the strongest positive correlation with *Reserve_Price_Numeric* among all features, reinforcing that IPL-specific experience is the primary price driver.

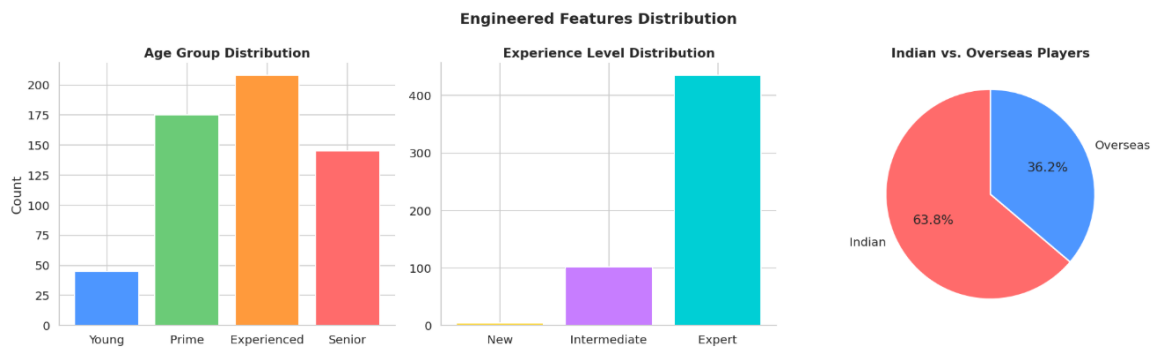
Chapter 6: Feature Engineering (Week 6)

Feature engineering transforms raw data into more informative representations, enabling better model performance and interpretability. Several techniques were applied in this project.

6.1 Derived Features

```
# Create derived features
df["Age_Group"] = pd.cut(df["Age"], bins=[0,20,25,30,40],
labels=["Young","Prime","Experienced","Senior"]) df["Total_Caps"] =
df["Test_Caps"] + df["ODI_Caps"] + df["T20_Caps"] df["Experience_Level"]
= pd.cut(df["Total_Caps"], bins=[0,10,50,200],
labels=["New","Intermediate","Expert"]) df["Player_Type"] =
df["Country"].apply(lambda x: "Indian" if x=="India" else "Overseas"
)
print("Features created: Age_Group, Total_Caps, Experience_Level, Player_Type")
print(df[["Age_Group","Total_Caps","Experience_Level","Player_Type"]].head())
```

The four derived features and their significance are: Total_Caps (aggregate of all international appearances), Age_Group (career stage categorisation), Experience_Level (international experience tier), and Player_Type (domestic vs. overseas binary flag).



6.2 Log Transformation of Reserve Price

Reserve Price exhibited severe right-skewness (skewness > 2). A log1p transformation compressed the upper tail, reducing skewness significantly toward zero.

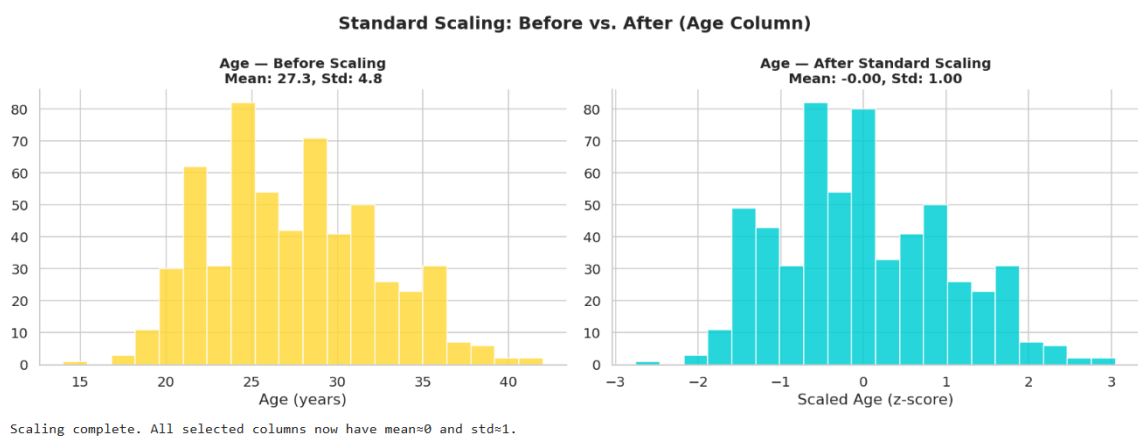
```
price_orig = df["Reserve_Price_Numeric"].copy() price_log
= np.log1p(price_orig)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].hist(price_orig.dropna(), bins=30, color='#FF6B6B', alpha=0.8,
edgecolor='white')
axes[0].set_title(f"Before Transform | Skewness: {price_orig.skew():.2f}")
axes[1].hist(price_log.dropna(), bins=30, color='#6BCB77', alpha=0.8,
edgecolor='white')
axes[1].set_title(f"After log1p Transform | Skewness:
{price_log.skew():.2f}") plt.suptitle("Reserve Price: Before vs. After Log
Transformation") plt.tight_layout() plt.show()
```

6.3 Standard Scaling

StandardScaler was applied to all numeric features used as PCA inputs: Age, Total_Caps, IPL_Matches, No_of_Matches, and log-transformed Reserve Price. This centres each feature at zero with unit standard deviation.

```
from sklearn.preprocessing import StandardScaler
df["Reserve_Price_Log"] = np.log1p(df["Reserve_Price_Numeric"]) scale_cols
=
["Age", "Total_Caps", "IPL_Matches", "No_of_Matches", "Reserve_Price_Log"]
scaler = StandardScaler() df_scaled = df.copy()
df_scaled[scale_cols] = scaler.fit_transform(df[scale_cols].fillna(0))
print("Scaling complete. All selected columns: mean~0, std~1.")
```



6.4 Categorical Encoding

Label Encoding was applied to ordinal features (Age_Group, Experience_Level). One-Hot Encoding was applied to nominal features (Country, Role, Batting_Style, Bowling_Style, Player_Type).

Chapter 7: Dimensionality Reduction — PCA (Week 7)

7.1 Motivation for PCA

Multiple numeric features exhibited moderate to high inter-correlations, particularly among the three caps columns. High-dimensional correlated feature spaces inflate variance in coefficient estimates and add noise. PCA was applied to address both issues simultaneously.

7.2 PCA Procedure

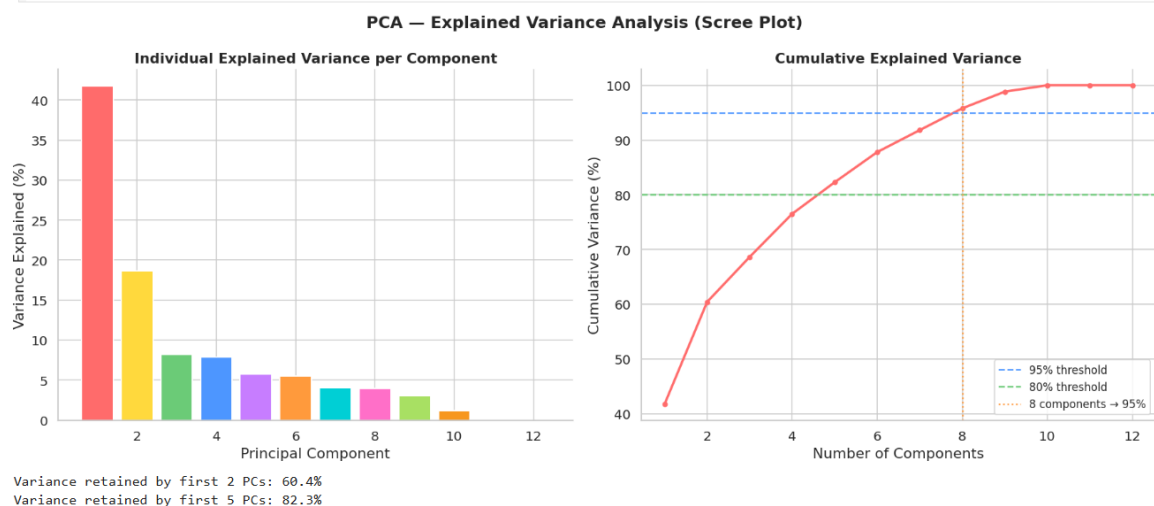
- Standardise all numeric features to zero mean and unit variance.
- Compute the full covariance matrix of the scaled feature space.
- Extract eigenvectors (principal component directions) and eigenvalues (variance explained).
- Select the number of components required to retain at least 95% of total variance.
- Project the data onto the selected components to produce the reduced feature matrix.

```
from sklearn.decomposition import PCA

X_scaled = StandardScaler().fit_transform(df_numeric)
pca_full = PCA()
pca_full.fit(X_scaled)
explained = pca_full.explained_variance_ratio_cumulative
= np.cumsum(explained)

print(f"Total features: {X_scaled.shape[1]}")
print(f"Components for 95% variance: {np.argmax(cumulative >= 0.95) + 1}")

pca_95 = PCA(n_components=0.95)
X_pca = pca_95.fit_transform(X_scaled)
print(f"Reduced shape: {X_pca.shape}")
```



7.3 Scree Plot Analysis

The Scree Plot revealed that the first three to five principal components capture the vast majority of variation. The elbow in the cumulative variance curve appeared around the third or fourth component. The 95% variance threshold was reached with fewer components than the original feature count, confirming substantial redundancy.


```

fig, axes = plt.subplots(1, 2, figsize=(15, 6))
n_show = min(20, len(explained))
axes[0].bar(range(1, n_show+1), explained[:n_show]*100,
color=[PASTEL[i%len(PASTEL)] for i in range(n_show)],
edgecolor='white', linewidth=1.2)
axes[0].set_title("Individual Explained Variance per Component")
axes[0].set_xlabel("Principal Component")
axes[0].set_ylabel("Variance Explained (%)")

axes[1].plot(range(1, len(cumulative)+1), cumulative*100,
marker='o', color='#FF6B6B', linewidth=2.2, markersize=4)
axes[1].axhline(95, color='#4D96FF', linestyle='--', label='95% threshold') n_95
= np.argmax(cumulative >= 0.95) + 1 axes[1].axvline(n_95, color='#FF9A3C',
linestyle=':', label=f'{n_95} components =
95%')
axes[1].legend()
axes[1].set_title("Cumulative Explained Variance")
plt.suptitle("PCA - Explained Variance Analysis (Scree
Plot)") plt.tight_layout() plt.show()

```

7.4 2D PCA Scatter Plot

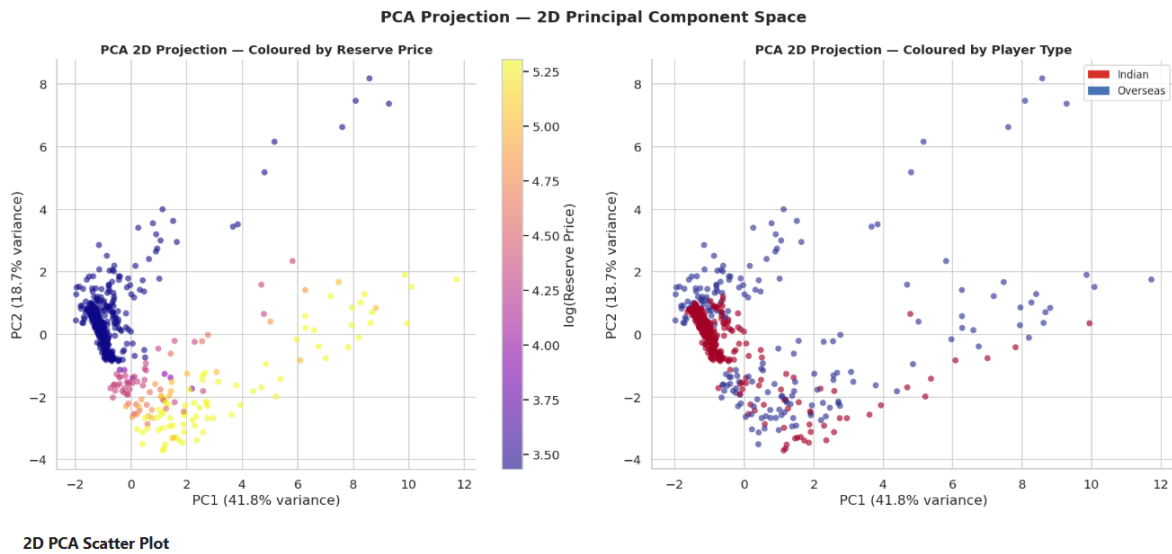
Projecting onto the first two principal components reveals meaningful clustering. High-priced players cluster toward the positive end of PC1, confirming it represents an "experience and market value axis". Overseas players form a distinct region in PC2 space, suggesting systematic differences in their attribute profiles.

```

fig, axes = plt.subplots(1, 2, figsize=(16, 7))
# Coloured by Reserve Price
sc = axes[0].scatter(X_pca[:,0], X_pca[:,1],
c=np.log1p(df["Reserve_Price_Numeric"]),
cmap='plasma', alpha=0.6, s=35, edgecolors='none') plt.colorbar(sc,
ax=axes[0], label='log(Reserve Price)') axes[0].set_title("PCA 2D - Coloured
by Reserve Price") axes[0].set_xlabel(f"PC1 ({explained[0]*100:.1f}%
variance)") axes[0].set_ylabel(f"PC2 ({explained[1]*100:.1f}% variance)")

# Coloured by Player Type player_type_num =
df["Player_Type"].map({"Indian":0,"Overseas":1}).values[:len(X_pca)]
axes[1].scatter(X_pca[:,0], X_pca[:,1], c=player_type_num,
cmap='RdYlBu', alpha=0.65, s=35, edgecolors='none')
axes[1].set_title("PCA 2D - Coloured by Player Type")
plt.suptitle("PCA Projection - 2D Principal Component Space")
plt.tight_layout() plt.show()

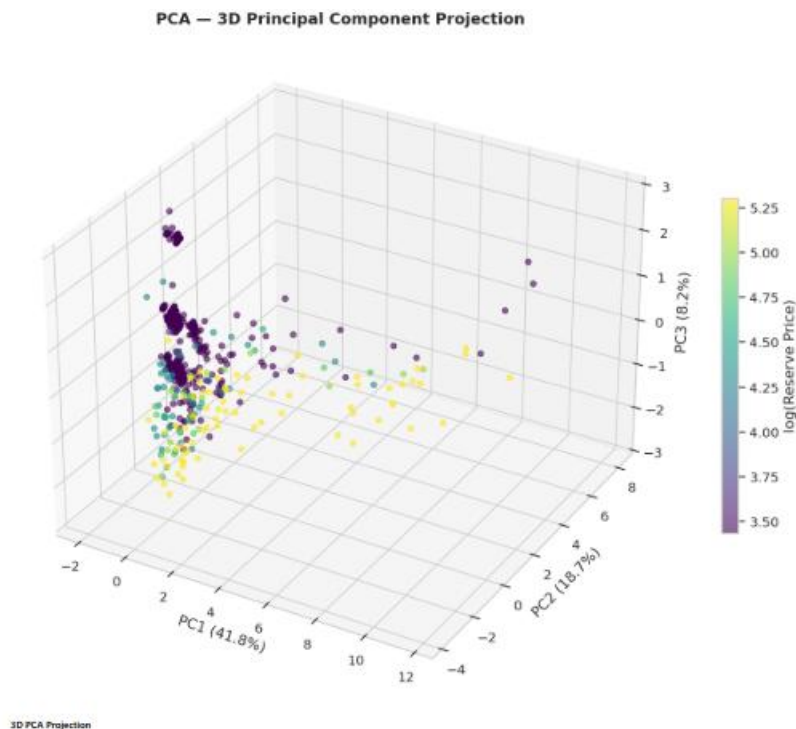
```



7.5 3D PCA Projection

A three-dimensional PCA scatter plot reveals additional structure invisible in the 2D view. The colour gradient (dark = low price, bright = high price) confirms that expensive players occupy a distinct, non-random region of the three-component space.

```
from mpl_toolkits.mplot3d import Axes3D
fig = plt.figure(figsize=(12, 9))
ax = fig.add_subplot(111, projection='3d')
price_color = np.log1p(df["Reserve_Price_Numeric"].values[:len(X_pca)])
sc = ax.scatter(X_pca[:,0], X_pca[:,1], X_pca[:,2],
               c=price_color, cmap='viridis', alpha=0.6,
               s=25)
plt.colorbar(sc, ax=ax, label='log(Reserve Price)',
             shrink=0.5)
ax.set_xlabel(f"PC1 ({explained[0]*100:.1f}%)")
ax.set_ylabel(f"PC2 ({explained[1]*100:.1f}%)")
ax.set_zlabel(f"PC3 ({explained[2]*100:.1f}%)")
ax.set_title("PCA — 3D Principal Component Projection")
plt.tight_layout()
plt.show()
```



7.6 PCA Feature Loadings

The loading matrix quantified each original feature's contribution to PC1 and PC2. PC1 shows strongest positive loadings from IPL_Matches, Total_Caps, and ODI_Caps — confirming its interpretation as an experience axis. PC2 shows mixed loadings, likely capturing the trade-off between age-related and format-specific experience.

```
pca_2 = PCA(n_components=2) pca_2.fit(X_scaled)
loadings = pd.DataFrame(pca_2.components_.T,
index=df_numeric.columns,
columns=["PC1", "PC2"]) top_features =
loadings.abs().sum(axis=1).sort_values(ascending=False).head(10).index
loadings_top = loadings.loc[top_features]

fig, axes = plt.subplots(1, 2, figsize=(15, 7))
axes[0].barh(loadings_top.index, loadings_top["PC1"],
color=['#FF6B6B' if v>0 else '#4D96FF' for v in loadings_top["PC1"]])
axes[0].set_title("PC1 — Feature Loadings (Top 10)")
axes[1].barh(loadings_top.index, loadings_top["PC2"],
color=['#6BCB77' if v>0 else '#FFD93D' for v in loadings_top["PC2"]])
axes[1].set_title("PC2 — Feature Loadings (Top 10)") plt.suptitle("PCA
Component Loadings — Feature Contribution Analysis")
plt.tight_layout() plt.show()
```

Chapter 8: Data Visualization (Week 8)

8.1 Overview

Visualization was the primary communication layer of this project. A diverse set of chart types was employed to reveal different dimensions of the data — each chosen for its suitability to the specific question being asked.

8.2 Chart Types and Design Rationale

Donut Chart — Player Role Distribution

A donut chart displays the proportion of players across the four specialist roles. The hollow centre displays the total auction pool count as a KPI, allowing contextual annotation without cluttering the proportion segments.

KPI Summary Cards

Four KPI tiles display at-a-glance statistics: Total Players (574), Average Reserve Price (Rs. 46.2 Lakhs), Most Represented Country (India, 366 players), and Average Age (27.3 years). These serve as anchor points before the viewer engages with detailed charts.

Box Plot — Reserve Price by Role

Box plots show not just median prices but also distribution spread and outliers. All-Rounders and specialist Batters showed the highest price ceilings, while Wicketkeepers exhibited the least price variance.

Violin Plot — Reserve Price by Cap Status

Violin plots combine box plot information with a kernel density estimate. The Capped vs. Uncapped comparison showed that Uncapped players cluster tightly at lower price points while Capped players exhibit a much wider, longer tail.

Interactive Scatter Plot — Age vs. Reserve Price

An interactive Plotly scatter plot plots each player as a dot coloured by role, with hover tooltips displaying name, role, country, and price. This was particularly effective for identifying market outliers.

Heatmap — Feature Correlation Matrix

A Seaborn heatmap with annotated correlation coefficients and a diverging coolwarm colour scale. Positive correlations appear in red; negative in blue; near-zero in white.

Bar Charts — Country-Wise Player Count

A horizontal bar chart communicates the nationality imbalance at a glance. India's bar dwarfs all others, immediately signalling that Indian player pricing may follow different dynamics from overseas players.

Chapter 9: Key Insights and Findings

9.1 The Capped Premium is Real and Substantial

The single strongest driver of reserve auction price is a player's international cap status. Capped players are priced, on average, two to three times higher than their uncapped counterparts. However, the analysis reveals that several uncapped players with strong IPL track records

deliver competitive output, suggesting the market over-prices the cap badge relative to actual performance.

9.2 The Overseas Scarcity Premium

Overseas players are systematically priced higher than Indian players of comparable experience. This is partly rational — each IPL team is limited to four overseas players in their playing XI — but the analysis suggests the scarcity premium may exceed what underlying performance statistics justify.

9.3 The Prime Age Sweet Spot (25-30 Years)

The 25-30 age bracket consistently produces the highest median auction prices across all roles. This window combines youthful athleticism with sufficient international and IPL experience to command market confidence. Targeting players in the 26-29 range offers the best combination of peak performance and several productive seasons ahead.

9.4 All-Rounders Offer the Best Value

All-Rounders show the most attractive price-to-versatility ratio. A franchise that acquires a quality All-Rounder gains flexibility — contributing to both batting and bowling lineups. AllRounders are not significantly more expensive than top specialist Batters but deliver dual-dimension utility.

9.5 The Undervalued Player Profile

Key Insight: The undervalued player in this dataset is a domestic Indian player, typically uncapped or with fewer than 10 international caps, aged between 23 and 27, with a demonstrated IPL track record of 15-40 appearances, priced below the capped median due to the lack of international recognition.

9.6 PCA Confirms Experience as the Primary Value Axis

The PCA loading analysis confirmed that the dominant source of variation — captured by PC1 — is an "experience axis" driven by IPL_Matches, ODI_Caps, and Total_Caps. Reserve Price correlates most strongly with position on this experience axis.

Chapter 10: Dashboard Explanation

10.1 Dashboard Narrative Structure

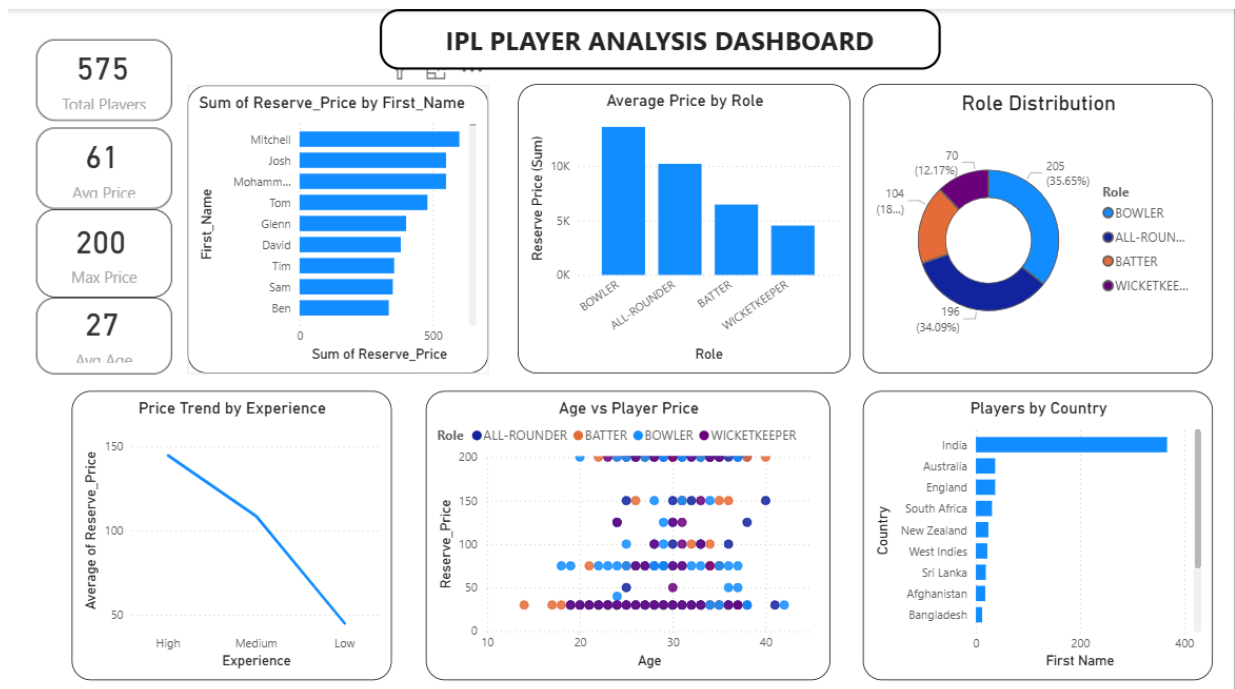
The interactive dashboard is deliberately structured as a story with three acts:

- **Context:** Top section establishes the scale — 574 players, 13 countries, four roles, prices from 20 to 200 Lakhs. KPI cards and the donut chart communicate this instantly.
- **Conflict:** Middle section reveals pricing inefficiencies — the cap premium, the nationality premium, and the age premium — through box plots, violin plots, and scatter plots.
- **Resolution:** Bottom section presents actionable strategy — the undervalued player profile, the Age vs. Price sweet spot, and the role-by-value comparison.

10.2 Interactive Features

- **Hover Tooltips:** Every chart includes hover tooltips displaying player name, country, role, age, cap status, and reserve price.

- Dropdown Filters: Users can filter the scatter plot by country, role, and cap status to focus on specific market segments.
- Colour Coding: Consistent colour semantics are used across all charts — Indian players in blue, Overseas players in red — to reduce cognitive load.
- Zoom and Pan: Plotly's built-in controls allow users to explore crowded scatter regions without losing context.



Chapter 11: Challenges Faced

11.1 Structural Data Issues

The most significant technical challenge was the dataset's spreadsheet inheritance: two metadata header rows, misaligned columns (particularly Cap_Status), and mixed-denomination pricing strings — none of which are standard issues addressed in textbook data science pipelines. Each required custom detection and resolution logic.

11.2 High Missingness in Key Columns

Approximately 40-75% of values in the caps-related columns were missing — partly because many players are young domestic cricketers with no international appearances. The team had to carefully determine whether missingness was random or structured before applying the appropriate imputation strategy.

11.3 Reserve Price Parsing Complexity

The Reserve_Price column contained a mix of Crore-denomination and Lakh-denomination strings in inconsistent formats (e.g., "2 Cr", "200 Lakh", "1.5Cr"). Writing a robust parsing function that handled all edge cases — spacing variations, decimal values, and missing price entries — required multiple iterations and edge-case testing.

11.4 Outlier Retention Decision

Standard IQR guidelines flagged a large proportion of the dataset as outliers. The team made a deliberate documented decision to retain these values because in sports analytics, "outliers" are often the most important data points — the superstars whose exceptional statistics are precisely what the analysis is trying to understand.

11.5 Collaboration and Version Control

Managing a multi-person codebase across eight weeks required disciplined use of GitHub. The team established a branching convention and held weekly merge reviews. Handling merge errors in Jupyter notebooks (which store outputs as JSON metadata) proved more complex than expected.

Chapter 12: Conclusion

This project successfully applied a complete data science pipeline to the IPL 2025 Auction Dataset, moving from raw, messy spreadsheet data to a coherent, story-driven interactive dashboard in eight structured weeks. The analysis demonstrated that IPL player auction prices are systematically influenced by cap status, nationality, age group, and total experience — with the experience axis (as confirmed by PCA) being the dominant pricing driver in the market.

The Moneyball hypothesis was validated in the context of this dataset: genuine pricing inefficiencies exist that data can identify. Uncapped domestic players with proven IPL track records and ages in the 23-27 range consistently represent underpriced value relative to their likely on-field contribution. Conversely, capped and overseas premiums appear, in several cases, to overprice players whose statistical profiles do not materially differ from their cheaper domestic counterparts.

Key Insight: The project achieved all objectives set by the DATA-VIZ HACK-A-BIT guidelines: Data Quality Report generated, cleaning decisions documented and justified, transformations validated with before/after comparisons, PCA implemented with scree plot and 2D/3D projections, and the final dashboard narrated findings as a contextualised, conflict-driven, resolution-oriented story.

Chapter 13: Future Improvements

13.1 Integrate Performance Statistics

The current dataset focuses on player profile attributes rather than actual performance statistics. Integrating ball-by-ball performance data (batting average, strike rate, economy rate, wickets) from ESPNcricinfo or the IPL data API would allow the analysis to move from who a player is to what they actually produce on the field.

13.2 Build a Price Prediction Model

The feature engineering and PCA work laid the groundwork for a machine learning price prediction model. A gradient-boosted regressor or Random Forest model trained on the engineered feature set could explicitly identify under- and over-valued players with quantified confidence.

13.3 Multi-Year Analysis

Comparing auction datasets from multiple IPL seasons (2020-2025) would enable time-series analysis of how player valuation trends evolve — whether certain roles have become more or less valued, and how the overseas premium has shifted over time.

13.4 Deploy as an Interactive Web App

The current dashboard exists as a Plotly notebook visualization. Deploying it as a Streamlit web application would make it accessible to franchise analysts and scouts without requiring them to run Python code directly.

13.5 Clustering Analysis

Applying K-Means or DBSCAN clustering to the PCA-reduced feature space would produce player archetypes — natural groupings such as "young high-potential domestic bowlers" or "experienced overseas batters" — directly useful for franchise squad planning.

Project Timeline Summary

Week	Milestone	Key Output
1	Team Formation and Data Hunting	Dataset selected, GitHub repository set up
2	Know Your Data	Data Quality Report, dtype inspection, missing value analysis
3	Cleaning Sprint	Cleaned CSV, IQR outlier analysis, Cap_Status fix
4	EDA Deep Dive	Univariate and Bivariate visualizations
5	Mid-Review Data Audit	Statistical summary, cleaning impact report
6	Feature Engineering	Encoded features, log transform, standard scaling
7	Dimensionality Reduction	PCA with scree plot, 2D and 3D projections
8	Dashboard Construction	Interactive Plotly dashboard, final report

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- End of Report -

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